Assignment No. 2

Q1) Hollow Square. n = 5

```
*  *  *  *  *
*           *
*           *
*           *
*  *  *  *  *
```

→

```
import java.util.*;
class Demo {
    psvm() {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 1; i ≤ n; i++) {
            for (int j = 1; j ≤ n; j++) {
                if (2 ≤ i ≤ n-1 && 2 ≤ j ≤ n-1) {
                    SOP(" _ ");
                }
```

```
        else {
            SOP("*");
        }
    }
    } SOPln();
}
}
```

Dry run:

i = 1 $1 \le 5$ i++

↓

j = 1 $1 \le 5$ j++

↓

if condition false

↳ * 2

$2 \le 5 \rightarrow$ false

↳ * 3

$3 \le 5 \rightarrow$ false

↳ * 4

$4 \le 5 \rightarrow$ false

↳ * 5

$5 \le 5 \rightarrow$ false

↳ * 6

$6 \nleq 5$

SOPln(); 2

i = 2 $2 \le 5$

↓

j = 1 $1 \le 5 \rightarrow$ false

↳ * 2

$2 \leq 5 \rightarrow$ true

↳ _ 3

$3 \leq 5 \rightarrow$ true

↳ _ 4

$4 \leq 5 \rightarrow$ true

↳ _ 5

$5 \leq 5 \rightarrow$ false

↳ * 6

$6 \nleq 5$ -

SOPln();

3

$3 \leq 5$

↓

j=1 $1 \leq 5 \rightarrow$ false

↳ * 2

$2 \leq 5 \rightarrow$ true

↳ _ 3

$3 \leq 5 \rightarrow$ true

↳ _ 4

$4 \leq 5 \rightarrow$ true

↳ _ 5

$5 \leq 5 \rightarrow$ false

↳ * 6

$6 \nleq 5$

SOPln();

4

Q2 Diagonal Square n=5

```
*       *
  *   *
    *
  *   *
*       *
```

```
import java.util.*;
class Demo{}
    psvm() {
        Scanner sc = new scanner(System.in);
        int n = sc.nextInt();
        for (int i=1; i≤n; i++) {
            for (int j=1; j≤n; j++){
                if (i==j || i+j==n+1){
                    sop(*);
                }
                else {
                    sop("-");
                }
            }
            sopln();
        }
    }
}
```

~~Prog run~~ :-

Dry run:-

$i=1 \quad 1 \le 5$

↓

$j=1 \quad 1 \le 5$

↳ if condition true

↳ * 2

$2 \le 5 \rightarrow$ false

↳ _ 3

$3 \le 5 \rightarrow$ false

↳ _ 4

$4 \le 5 \rightarrow$ false

↳ _ 5

$5 \le 5 \rightarrow$ ~~false~~ true

↳ * 6

$6 \nleq 5$

sop ln(); 2

$2 \le 5$

↓

$j=1 \quad 1 \le 5 \rightarrow$ false

↳ _ 2

$2 \le 5 \rightarrow$ true

↳ * 3

$3 \le 5 \rightarrow$ ~~true~~ false

↳ _ 4

$4 \le 5 \rightarrow$ ~~false~~ true

↳ * 5

$5 \le 5 \rightarrow$ ~~false~~ false

↳ _ 6

$6 \nleq 5$

sop ln(); 3

$3 \leq 5$

↓

$j = 1$ $\quad 1 \leq 5 \rightarrow$ false

↳ _ $\quad 2$

$2 \leq 5 \rightarrow$ false

↳ _ $\quad 3$

$3 \leq 5 \rightarrow$ true

↳ * $\quad 4$

$4 \leq 5 \rightarrow$ false

~~5 ≤ 5~~ → _ $\quad 5$

$5 \leq 5 \rightarrow$ false

↳ _ $\quad 6$

$6 \nleq 5$

SOPln();

4

Q3) Right Triangle $n=5$

```
*
* *
* * *
* * * *
* * * * *
```

```
import java.util.*;
class Demo{
  psvm(){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    for (int i=1; i≤n; i++){
      for (int j=1; j≤n; j++){
        if (j <= i){
          SOP("*");
        }
      }
    SOPln();
  }
```

```
for (int i=1; i≤n; i++){
    for (int j=1; j≤i; j++){
        SOP ('*');
    }
    SOPln();
}
```

i=1 i ≤ n i++

1 ≤ 5

↓

j=1 j ≤ i j++

1 ≤ 1 → true

↳ * 2

2 ≤ 1

SOPln(); 2

2 ≤ 5

↓

j=1 1 ≤ 2 → true

↳ * 2

2 ≤ 2 → true

↳ *

3 ≤ 2 → false

SOPln(); 3

3 ≤ 5

↓

j=1 1 ≤ 2 → true

↳ * 2

2 ≤ 2 → true

↳ * 3

3 ≤ 3 → true

↳ * 4

4 ≤ 3 → false

Q4 Inverted right-angled triangle

```
* * * * *
* * * *
* * *
* *
*
```

→

```
Class Demo{
    psvm() {
        int n = 5;
        for (int i = 1; i <= n; i++){
            for (int j = 1; j <= n; j++){
                if (i + j <= 6){
                    sop("*");
                }else {
                    sop("_");
                }
            }
            sopln();
        }
    }
}
```

```
for (int i = 1; i <= n; i++){
    for (int j = 1; j <= n-i+1; j++){
        sop("*");
    }
    sopln();
}
```

Dry run: for n = 5

i = 1 1 ≤ 5

↓

j = 1 1 ≤ 5 - 1 + 1 j++

↳ * 2

2 ≤ 5 - 1 + 1

↳ * 3

3 ≤ 5 - 1 + 1

↳ * 4

4 ≤ 5 - 1 + 1

↳ * 5

5 ≤ 5 - 1 + 1

↳ * 6

6 ≰ 5

Sopln(); → * * * * *

i = 2 2 ≤ 5

↓

j = 1 1 ≤ 5 - 2 + 1 j++

↳ * 2

2 ≤ 5 - 2 + 1

↳ * 3

3 ≤ 5 - 2 + 1

↳ * 4

4 ≤ 5 - 2 + 1

↳ * 5

5 ≰ 5 - 2 + 1

Sopln(); → * * * *

Q5]

```
* # * #
# * # *
* # * #
# * # *
```

```
class Demo {
    psvm() {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 1; i ≤ n; i++) {
            for (int j = 1; j ≤ n; j++) {
                if ((i+j) % 2 == 0) {
                    SOP("*");
                } else {
                    SOP("#");
                }
            }
            SOPln();
        }
    }
```

```
boolean check = true;
for (int i = 1; i ≤ n; i++) {
    for (int j = 1; j ≤ n; j++) {   → check = !check
        if (check) {
            SOP("*");
            check = false;
        } else {
            SOP("#");
            check = true;
        }
    }
}
```

My run for n = 4

i = 1 $1 \le 4$

↓

j = 1 $1 \le 4$ → if true j++

↳ * 2

$2 \le 4$ → if false

↳ # 3

$3 \le 4$ → if true

↳ * 4

$4 \le 4$ if false

↳ # 5

$5 \nleq 4$ → * # * #

SoPln();

i = 2 $2 \le 4$

↓

j = 1 $1 \le 4$ → if false j++

↳ # 2

$2 \le 4$ → if true

↳ * 3

$3 \le 4$ → if false

↳ # 4

$4 \le 4$ → if true

↳ * 5

$5 \nleq 4$

SoPln(); → # * # *